

## AGS Vic-Ring Systems

### STYLE W07 – Rigid Coupling

The Style W07 AGS (Advanced Groove System) coupling is comprised of two housings and is designed to provide a strong connection for large diameter pressure piping systems. It is the first flat pad, metal-to-metal, rigid coupling to be offered within this size range. Its unique wedge-shaped key profile results in easier initial assembly alignment. The Style W07 AGS coupling in 26 – 50"/650 – 1250 mm sizes, features lifting lugs integrated into the housings to help ease handling during installation.

The Style W07 coupling is mounted onto an external ring utilizing the AGS groove profile. The *Vic-Ring* system provided by Victaulic maintains full pipe wall thickness for slurries, tailings, and a variety of other services.

The Style W07 AGS *Vic-Ring* system is available for pipe sizes ranging from 12 – 48"/300 – 1200 mm. The coupling used to join these pipe sizes will always be the next largest size (ex: 12"/300 mm pipe requires the use of a 14"/350 mm coupling). The larger size is needed to allow for the extra outside pipe diameter the AGS ring adds to the pipe.

Style W07 AGS couplings are rated up to 350 psi/2400 kPa for 14 – 24"/350 – 600 mm piping systems, to 300 psi/2065 kPa for 26 – 42"/650 – 1050 mm, and to 232 psi/1600 kPa for 44 – 50"/1100 – 1250 mm.

Style W07 AGS couplings are provided with FlushSeal® gaskets for a variety of services. Please specify gasket grade when ordering. Please refer to publication 05.01 for gasket service ratings.

Flexible couplings for *Vic-Ring* systems are also available. Refer to publication 16.12.



12 – 22"/300 – 550 MM SIZES



24 – 48"/600 – 1200 MM SIZES

#### WARNING



#### WARNING

- Victaulic AGS couplings **MUST ONLY** be installed on AGS *Vic-Ring* systems provided by Victaulic.

Failure to follow these instructions will cause grooves that are not within Victaulic AGS specifications, resulting in joint failure, serious personal injury, and property damage.

| Pipe OD Requirements* |                                     |
|-----------------------|-------------------------------------|
| Nominal Pipe Size     | Maximum Pipe OD                     |
| In. / mm              | In. / mm                            |
| 12<br>300             | 12.88<br>327                        |
| 14<br>350             | 14.14<br>359                        |
| 16<br>400             | 16.16<br>410                        |
| 18<br>450             | 18.18<br>462                        |
| 20<br>500             | 20.20<br>513                        |
| 24<br>600             | 24.25<br>616                        |
| 26 – 48<br>650 – 1200 | Nominal Pipe Size<br>+ 0.250"/6.4mm |

\* For pipe exceeding these values contact Victaulic.

#### JOB/OWNER

System No. \_\_\_\_\_

Location \_\_\_\_\_

#### CONTRACTOR

Submitted By \_\_\_\_\_

Date \_\_\_\_\_

#### ENGINEER

Spec Sect \_\_\_\_\_ Para \_\_\_\_\_

Approved \_\_\_\_\_

Date \_\_\_\_\_

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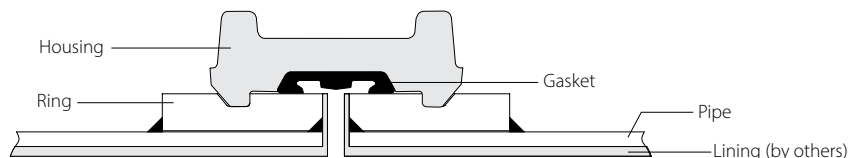


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## AGS Vic-Ring Systems

### STYLE W07 – Rigid Coupling

#### MATERIAL SPECIFICATIONS



#### **Vic-Rings:** Victaulic Type “D” Vic-Rings

- **Material:**

Carbon Steel to ASTM A 105 or equivalent

\* Optional: ASTM A105N (Specify at time of order)

Other materials available upon request contact Victaulic for details.

**Housing:** Ductile iron conforming to ASTM A-536, grade 65-45-12. Ductile iron conforming to ASTM A-395, grade 65-45-15, is available upon special request.

**Housing Coating:** Orange enamel.

- Optional: Hot dipped galvanized and others.

#### **Coupling Gasket:** (specify choice\*)

- **Grade “E” FlushSeal EPDM**

EPDM (Green color code). Temperature range –30°F to +230°F / –34°C to +110°C.

Recommended for cold and hot water service within the specified temperature range plus a variety of dilute acids, oil-free air and many chemical services. UL classified in accordance with ANSI/NSF 61 for cold +86°F/+30°C and hot +180°F/+82°C potable water service. NOT RECOMMENDED FOR PETROLEUM SERVICES.

- **Grade “T” FlushSeal nitrile**

Nitrile (Orange color code). Temperature range –20°F to +180°F / –29°C to +82°C.

Recommended for petroleum products, air with oil vapors, vegetable and mineral oils within the specified temperature range. Not recommended for hot water services over +150°F/+66°C or for hot dry air over +140°F/+60°C.

- **Grade “L” FlushSeal silicone**

Silicone (Red color code). Temperature range –30°F to +350°F / –34°C to +177°C.

Recommended for dry heat, air without hydrocarbons to +350°F/+177°C and certain chemical services.

\* Services listed are General Service Recommendations only. It should be noted that there are services for which these gaskets are not recommended. Reference should always be made to the latest Victaulic Gasket Selection Guide for specific gasket service recommendations and for a listing of services which are not recommended.

**Bolts:** Heat-treated plated carbon steel, trackhead meeting the physical and chemical requirements of ASTM A-449 and physical requirements of ASTM A-183.

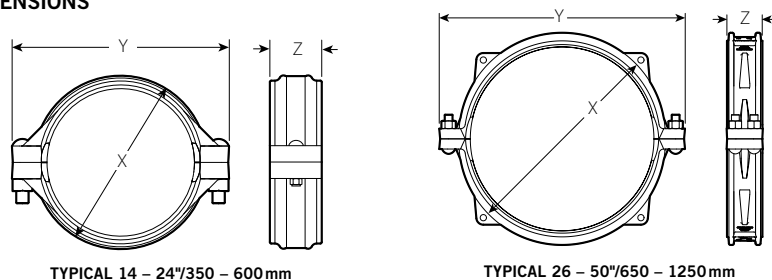
**Nuts:** Plated carbon steel, heavy hex conforming to ASTM A-563, Grade B.

**Washers:** Plated carbon steel, flat, SAE high strength conforming to ASTM F-436

## AGS Vic-Ring Systems

### STYLE W07 – Rigid Coupling

#### DIMENSIONS



TYPICAL 14 – 24"/350 – 600mm

TYPICAL 26 – 50"/650 – 1250mm

| Size                              |                               | Max. Working Pressure* | Max. End Load*    | Allow. Pipe End Sep. | Bolt/Nut No - Size | Dimensions – Inches/mm |               |              | Approx. Wgt. Each           |
|-----------------------------------|-------------------------------|------------------------|-------------------|----------------------|--------------------|------------------------|---------------|--------------|-----------------------------|
| Nominal Pipe Size<br>Inches<br>mm | Coupling Size<br>Inches<br>mm | psi/kPa                | Lbs./N            | Inches<br>mm         | Inches             | X                      | Y             | Z            | Coupling Only<br>Lbs.<br>kg |
| 12<br>300                         | 14.000<br>355.6               | 350<br>2500            | 53900<br>239757   | 0.25<br>6.4          | 2 – 1 x 5 ½        | 15.88<br>403           | 20.63<br>524  | 4.75<br>121  | 49.0<br>22.2                |
| 14<br>350                         | 16.000<br>406.4               | 350<br>2500            | 70400<br>313155   | 0.25<br>6.4          | 2 – 1 x 5 ½        | 18.13<br>460           | 23.50<br>597  | 4.75<br>121  | 61.0<br>27.7                |
| 16<br>400                         | 18.000<br>457.0               | 350<br>2500            | 89100<br>396335   | 0.25<br>6.4          | 2 – 1 x 5 ½        | 20.25<br>514           | 25.50<br>648  | 4.75<br>121  | 71.0<br>32.2                |
| 18<br>450                         | 20.000<br>508.0               | 350<br>2500            | 109950<br>489304  | 0.25<br>6.4          | 2 – 1 ½ x 5 ½      | 22.50<br>572           | 27.13<br>689  | 4.75<br>121  | 82.0<br>37.2                |
| 20<br>500                         | 22.000<br>558.8               | 350<br>2500            | 133046<br>592055  | 0.25<br>6.4          | 2 – 1 ½ x 6        | 24.75<br>628           | 29.25<br>743  | 4.75<br>121  | 99.0<br>44.9                |
| 22<br>550                         | 24.000<br>610.0               | 350<br>2500            | 158336<br>704275  | 0.25<br>6.4          | 2 – 1 ½ x 5 ½      | 26.63<br>677           | 32.38<br>822  | 4.75<br>121  | 116.0<br>52.6               |
| 24<br>600                         | 26.000<br>660.4               | 300<br>2065            | 159279<br>708508  | 0.38<br>9.6          | 4 – 1 ½ x 6        | 30.13<br>765           | 35.25<br>895  | 5.75<br>146  | 205.0<br>93.0               |
| 26<br>650                         | 28.000<br>711.2               | 300<br>2065            | 184726<br>821702  | 0.38<br>9.6          | 4 – 1 ½ x 6        | 32.25<br>819           | 37.25<br>945  | 5.75<br>146  | 220.0<br>99.8               |
| 28<br>700                         | 30.000<br>762.0               | 300<br>2065            | 212058<br>943281  | 0.38<br>9.6          | 4 – 1 ¼ x 7        | 33.88<br>863           | 39.63<br>1007 | 5.75<br>146  | 227.0<br>103.0              |
| 30<br>750                         | 32.000<br>812.8               | 300<br>2065            | 241274<br>1073240 | 0.38<br>9.6          | 4 – 1 ¼ x 7        | 36.13<br>918           | 41.75<br>1060 | 5.75<br>146  | 242.0<br>109.8              |
| 32<br>800                         | 34.000<br>865                 | 300<br>2065            | 272375<br>1212070 | 0.38<br>9.6          | 4 – 1 ¼ x 7        | 38.25<br>972           | 43.75<br>1111 | 5.75<br>146  | 255.0<br>115.7              |
| 34<br>850                         | 36.000<br>914.4               | 300<br>2065            | 340235<br>1513365 | 0.38<br>9.6          | 4 – 1 ¼ x 7        | 40.25<br>1022          | 45.75<br>1161 | 5.75<br>146  | 268.0<br>121.6              |
| 36<br>900                         | 38.000<br>965.2               | 300<br>2065            | 305363<br>1358322 | 0.38<br>9.6          | 4 – 1 ¼ x 7        | 41.25<br>1048          | 48.00<br>1219 | 5.75<br>146  | 298.0<br>135.0              |
| 38<br>950                         | 40.000<br>1016.0              | 300<br>2065            | 376991<br>1676940 | 0.44<br>11.1         | 4 – 1 ½ x 8        | 44.00<br>1117          | 50.50<br>1283 | 6.50<br>165  | 340.0<br>154.2              |
| 40<br>1000                        | 42.000<br>1066.8              | 300<br>2065            | 415632<br>1848823 | 0.44<br>11.1         | 4 – 1 ½ x 8        | 46.00<br>1168          | 52.50<br>1334 | 6.50<br>165  | 360.0<br>163.3              |
| 42<br>1050                        | 44.000<br>1150                | 232<br>1600            | 352763<br>1569795 | 0.44<br>11.1         | 4 – 1 ½ x 8        | 48.00<br>1219          | 54.50<br>1384 | 6.50<br>165  | 390.0<br>176.9              |
| 44<br>1100                        | 46.000<br>1168.4              | 232<br>1600            | 385561<br>1715746 | 0.44<br>11.1         | 4 – 1 ½ x 8        | 50.25<br>1277          | 56.50<br>1435 | 6.50<br>165  | 415.0<br>188.2              |
| 46<br>1150                        | 48.000<br>1219.2              | 232<br>1600            | 419820<br>1868199 | 0.44<br>11.1         | 4 – 1 ½ x 8        | 52.25<br>1328          | 58.50<br>1485 | 6.50<br>165  | 425.0<br>192.8              |
| 48<br>1200                        | 50.000<br>1270.0              | 232<br>1600            | 455530<br>2026298 | 0.44<br>11.1         | 4 – 1 ½ x 8        | 53.63<br>1362          | 60.88<br>1546 | 10.00<br>254 | 500.0<br>227.0              |

\* **Working Pressure** and **End Load** are total, from all internal and external loads, based on carbon steel, AGS rings installed in accordance with Victaulic® specifications. Contact Victaulic for performance on other ring materials.

WARNING: FOR ONE TIME FIELD TEST ONLY, the Maximum Joint Working Pressure may be increased to 1 ½ times the figures shown.

Style W07 AGS couplings are essentially rigid and do not permit expansion/contraction.

NOTE: Metric thread size bolts are available (color coded gold) for all coupling sizes upon request. Contact Victaulic for details.

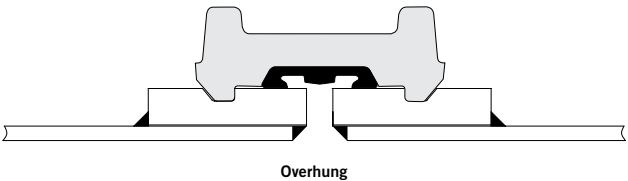
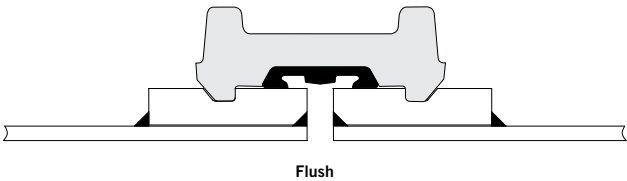
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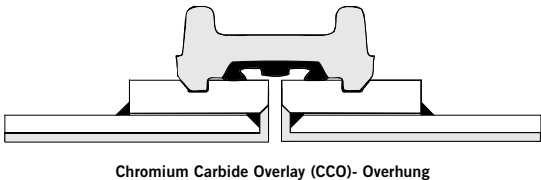
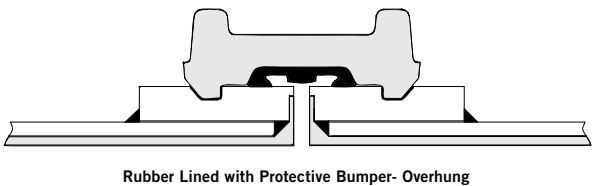
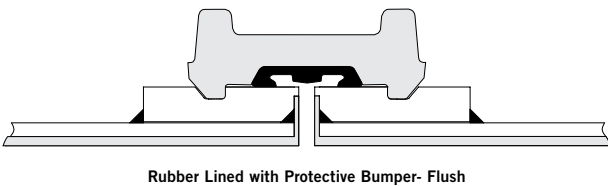
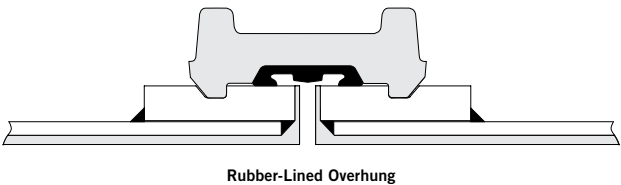
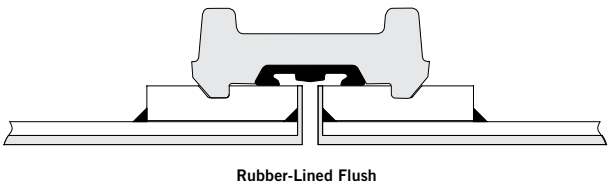
STYLE W07 – Rigid Coupling

LINING OPTIONS

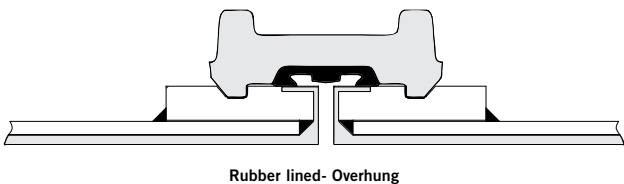
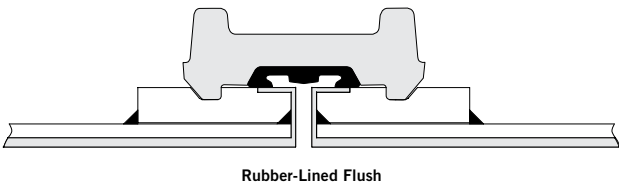
UNLINED SERVICES\*



ABRASION SERVICES



ABRASION AND CORROSION SERVICES

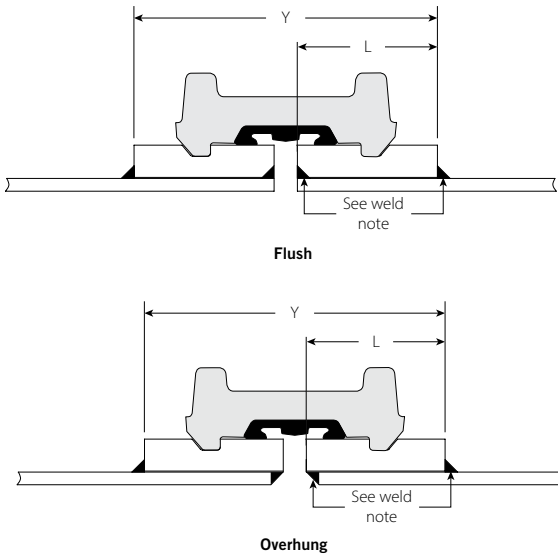


\*Suitable for use on abrasive and non-abrasive services

# AGS Vic-Ring Systems

## STYLE W07 – Rigid Coupling

### ALL SERVICE OPTIONS



| Size                        |                               |                           | AGS Vic-Ring  |               |                                     |
|-----------------------------|-------------------------------|---------------------------|---------------|---------------|-------------------------------------|
| Nominal Pipe Size<br>In./mm | Coupling Size<br>Inches<br>mm | Ring O.D.<br>Inches<br>mm | L†<br>In./mm  | Y†<br>In./mm  | Approx. Ring Weight Each<br>Lbs./kg |
| 12<br>300                   | 14.000<br>355.6               | 14.000<br>355.6           | 3.00<br>76.2  | 6.25<br>158.8 | 19.0<br>8.6                         |
| 14<br>350                   | 16.000<br>406.4               | 16.000<br>406.4           | 3.00<br>76.2  | 6.25<br>158.8 | 37.0<br>16.8                        |
| 16<br>400                   | 18.000<br>457.0               | 18.000<br>457.0           | 3.00<br>76.2  | 6.25<br>158.8 | 41.0<br>18.6                        |
| 18<br>450                   | 20.000<br>508.0               | 20.000<br>508.0           | 3.00<br>76.2  | 6.25<br>158.8 | 46.0<br>20.9                        |
| 20<br>500                   | 22.000<br>558.8               | 22.000<br>558.8           | 3.00<br>76.2  | 6.25<br>158.8 | 51.0<br>23.1                        |
| 22<br>550                   | 24.000<br>610.0               | 24.000<br>610.0           | 3.00<br>76.2  | 6.25<br>158.8 | 56.0<br>25.4                        |
| 24<br>600                   | 26.000<br>660.4               | 26.000<br>660.4           | 4.00<br>101.6 | 8.38<br>212.9 | 81.0<br>36.7                        |
| 26<br>650                   | 28.000<br>711.2               | 28.000<br>711.2           | 4.00<br>101.6 | 8.38<br>212.9 | 87.0<br>39.5                        |
| 28<br>700                   | 30.000<br>762.0               | 30.000<br>762.0           | 4.00<br>101.6 | 8.38<br>212.9 | 94.0<br>42.6                        |
| 30<br>750                   | 32.000<br>812.8               | 32.000<br>812.8           | 4.00<br>101.6 | 8.38<br>212.9 | 100.0<br>48.4                       |
| 32<br>800                   | 34.000<br>865                 | 34.000<br>865             | 4.00<br>101.6 | 8.38<br>212.9 | 106.0<br>48.1                       |
| 34<br>850                   | 36.000<br>914.4               | 36.000<br>914.4           | 4.00<br>101.6 | 8.38<br>212.9 | 113.0<br>51.3                       |
| 36<br>900                   | 38.000<br>965.2               | 38.000<br>965.2           | 4.00<br>101.6 | 8.38<br>212.9 | 120.0<br>54.4                       |
| 38<br>950                   | 40.000<br>1016.0              | 40.000<br>1016.0          | 4.25<br>108.0 | 8.94<br>227.1 | 133.0<br>60.3                       |
| 40<br>1000                  | 42.000<br>1066.8              | 42.000<br>1066.8          | 4.25<br>108.0 | 8.94<br>227.1 | 140.0<br>63.5                       |
| 42<br>1050                  | 44.000<br>1150                | 44.000<br>1150            | 4.25<br>108.0 | 8.94<br>227.1 | 146.0<br>66.2                       |
| 44<br>1100                  | 46.000<br>1168.4              | 46.000<br>1168.4          | 4.25<br>108.0 | 8.94<br>227.1 | 152.0<br>69.0                       |
| 46<br>1150                  | 48.000<br>1219.2              | 48.000<br>1219.2          | 4.25<br>108.0 | 8.94<br>227.1 | 160.0<br>72.6                       |
| 48<br>1200                  | 50.000<br>1270                | 50.000<br>1270            | 4.25<br>108.0 | 8.94<br>227.1 | 186.0<br>84.4                       |

Weld Note: Minimum recommended weld size is 3/8"/9.7 mm for 14"/350 mm and larger standard and extra heavy pipe. For 12"/300 mm pipe the minimum recommended weld size is 1/4"/6.4 mm

† For allowable pipe end separation see dimensional table on page 3.

‡ "L" dimension is from inside face to end of ring for all lined and un-lined configurations.

## AGS Vic-Ring Systems

### STYLE W07 – Rigid Coupling

#### TORQUE REQUIREMENTS

| Size                                                         | Recommended Torque |
|--------------------------------------------------------------|--------------------|
| Nominal Pipe Size<br>Inches<br>mm                            | ft. lbs.<br>N•m    |
| 14, 16, 18<br>350, 400, 450                                  | 250<br>340         |
| 20, 22, 24, 26, 28<br>500, 550, 600, 650, 700                | 375<br>500         |
| 30, 32, 34, 36, 38<br>750, 800, 850, 900, 950                | 500<br>680         |
| 40, 42, 44, 46, 48, 50<br>1000, 1050, 1100, 1150, 1200, 1250 | 600<br>813         |

#### INSTALLATION

Reference should always be made to the Victaulic Field Installation Instructions for the product you are installing. Victaulic Field Installation Instructions are included with each shipment of Victaulic products for complete installation and assembly data, and are available in PDF format on our website at [www.victaulic.com](http://www.victaulic.com).

#### WARRANTY

Refer to the Warranty section of the current Price List or contact Victaulic for details.

#### NOTE

This product shall be manufactured by Victaulic or to Victaulic specifications. All products to be installed in accordance with current Victaulic installation/assembly instructions. Victaulic reserves the right to change product specifications, designs and standard equipment without notice and without incurring obligations.

For complete contact information, visit [www.victaulic.com](http://www.victaulic.com)

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